

Analysis group meeting

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10/15/24



MISSISSIPPI STATE
UNIVERSITY™

Cooking chef

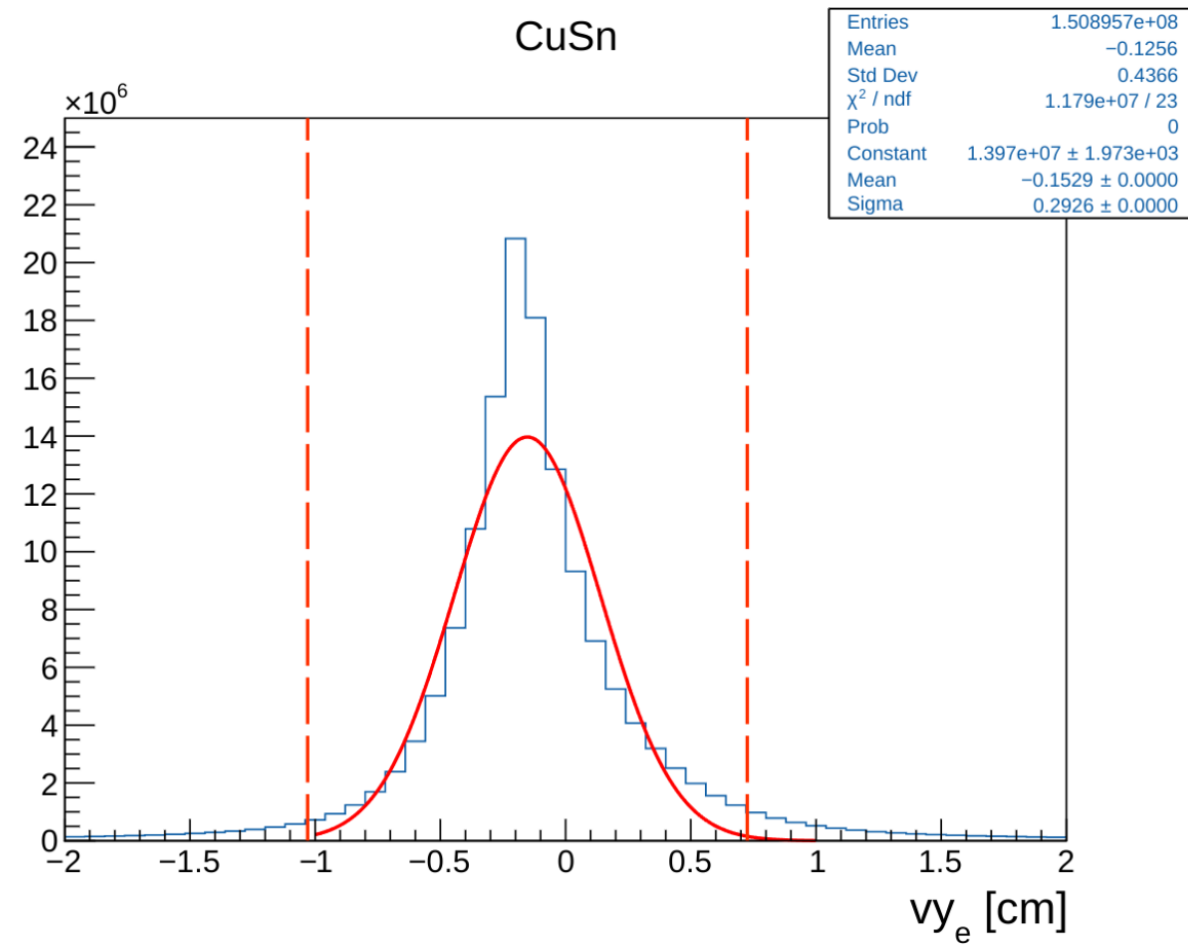
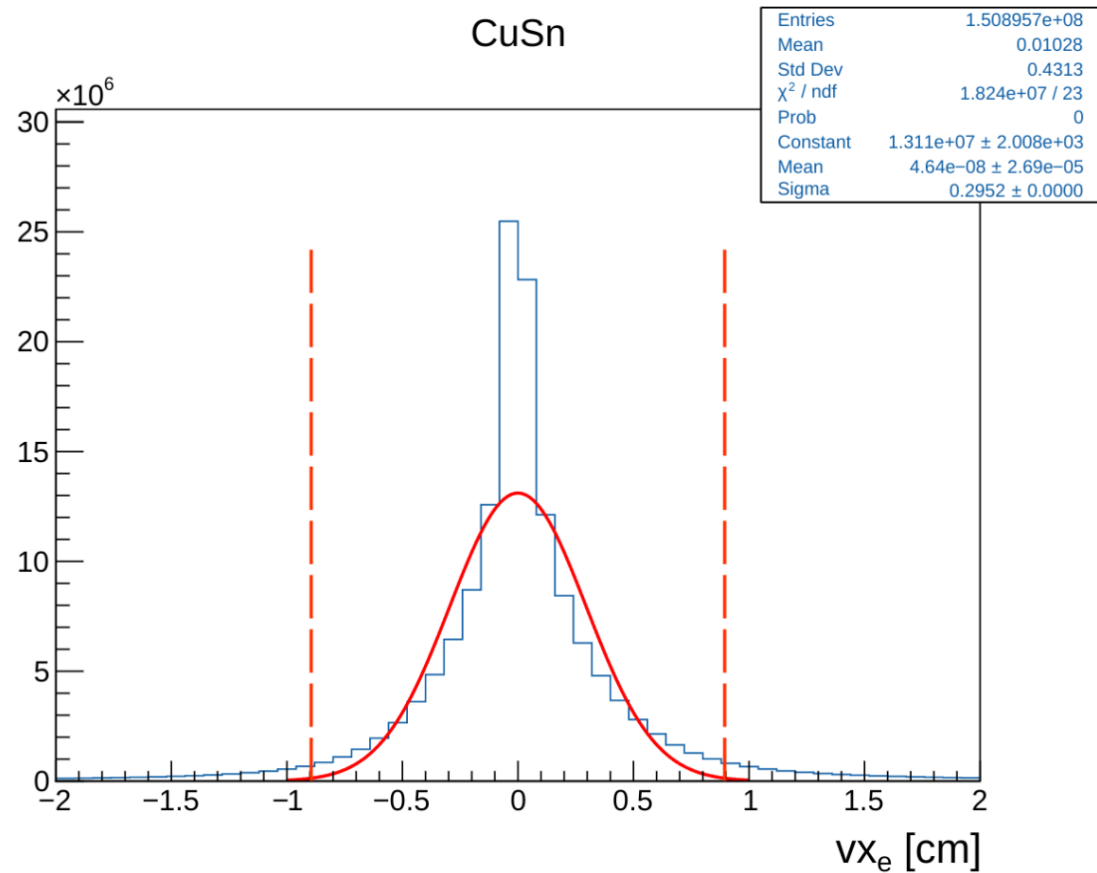
- Start cooking for pass0v5 after first FTOF calibration.
- Reconstruction have start CuSn, CxC, LD2 and ShortEmpRand.
- But no histograms.
- Change link for the yaml files on the wiki/Data Processing to point to <https://github.com/mathieuouillon/rgd-yaml>.

rgd-drqa-pass0v5_018316-18316	83.33%	24	2	2	0	0	0	0	0	0
rgd-drqa-pass0v5_ShortEmpRand-18305x45	71.66%	1062	116	185	0	171	12	0	0	0
rgd-drqa-pass0v5_CuSn-18347x301	74.28%	7224	738	1120	0	1112	2	1	0	0
rgd-drqa-pass0v5_CxC-18339x125	68.93%	3000	359	573	1	355	194	7	0	0
rgd-drqa-pass0v5_LD2-18307x147	66.24%	3528	434	757	1	426	252	66	0	0

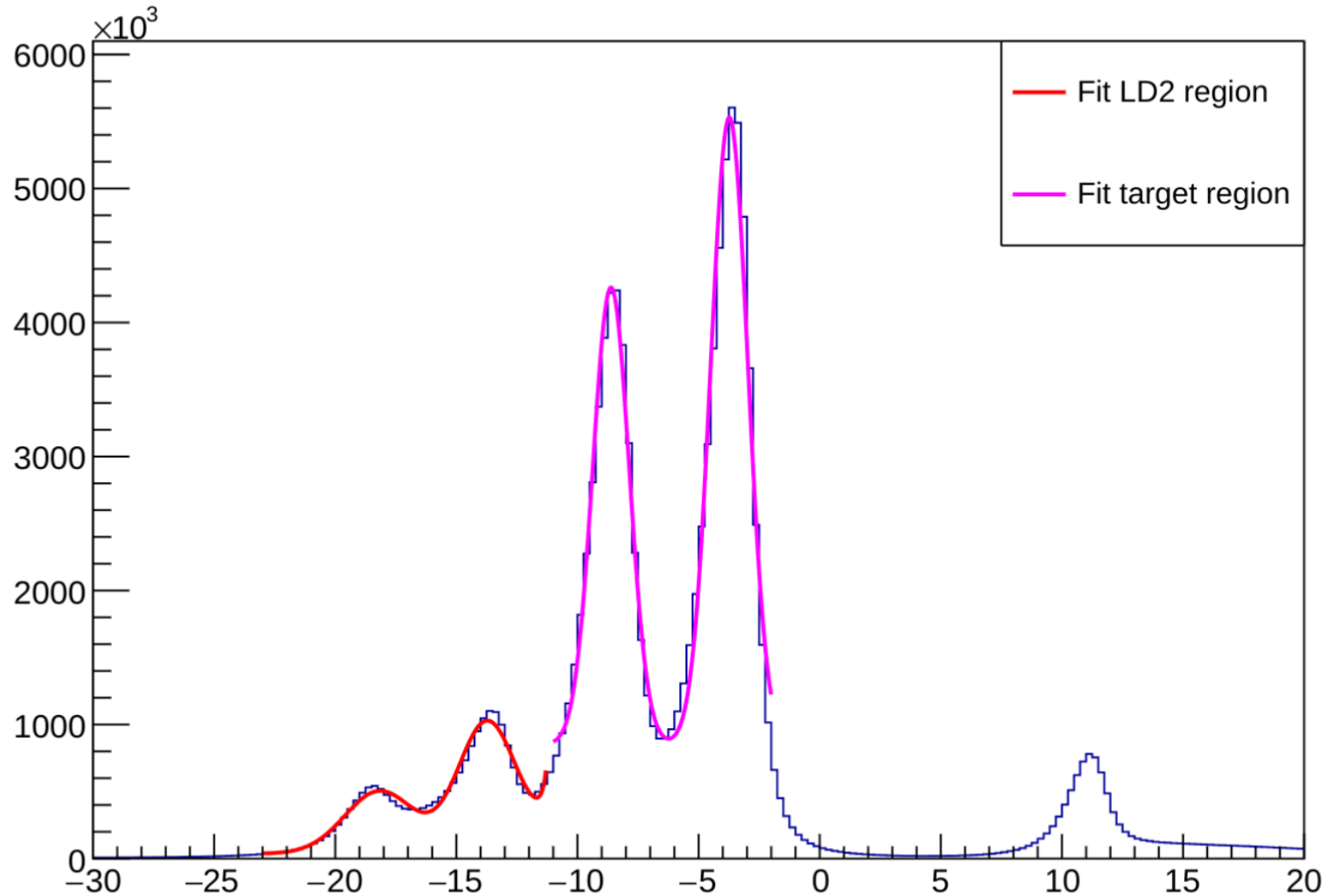
RGD analysis: vz study

Data: CuSn, pass0v4 all available

Cut on vx and vy at 3σ



RGD analysis: vz study

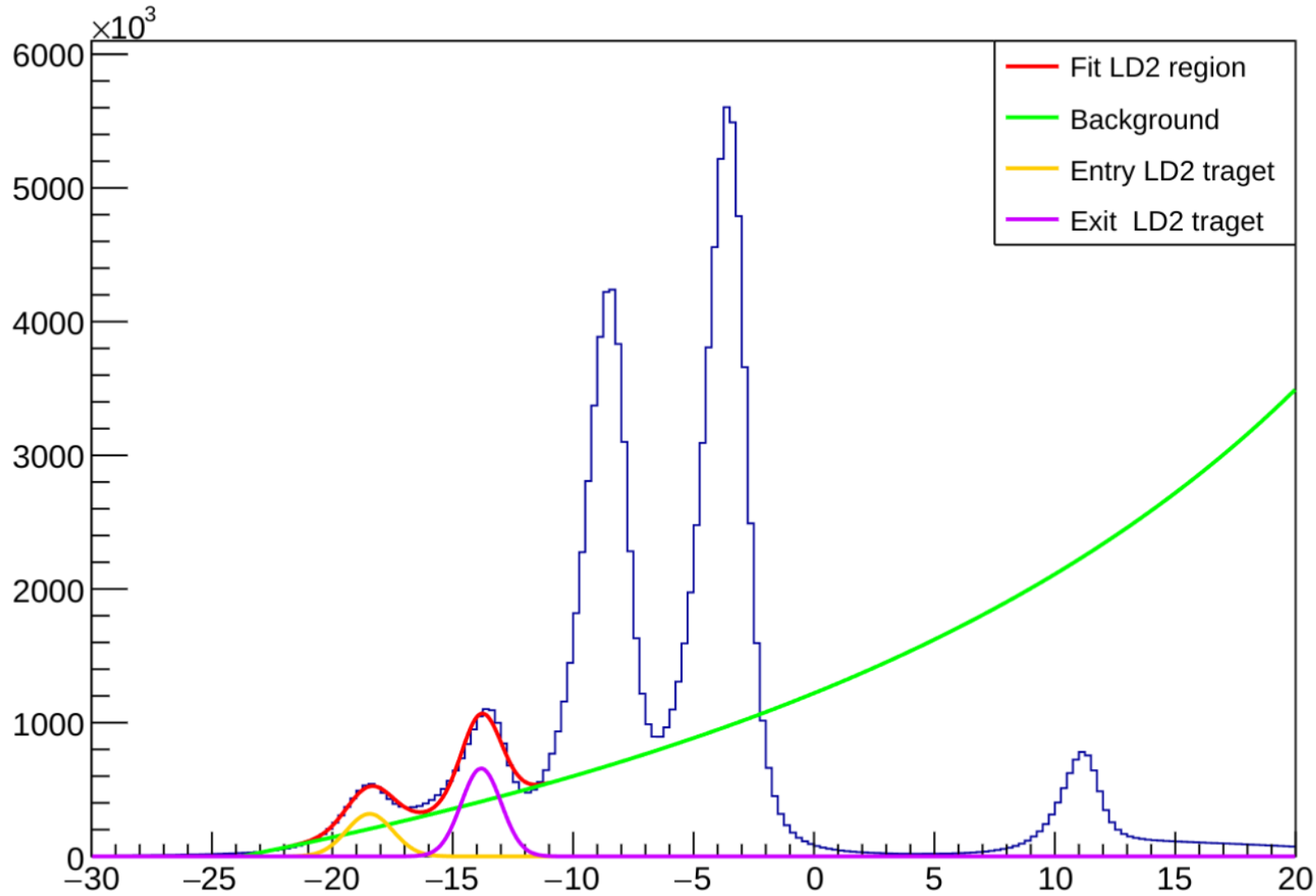


Data: CuSn, pass0v4 all available

Fit LD2 region with $1/x + \text{gaus} + \text{gaus}$

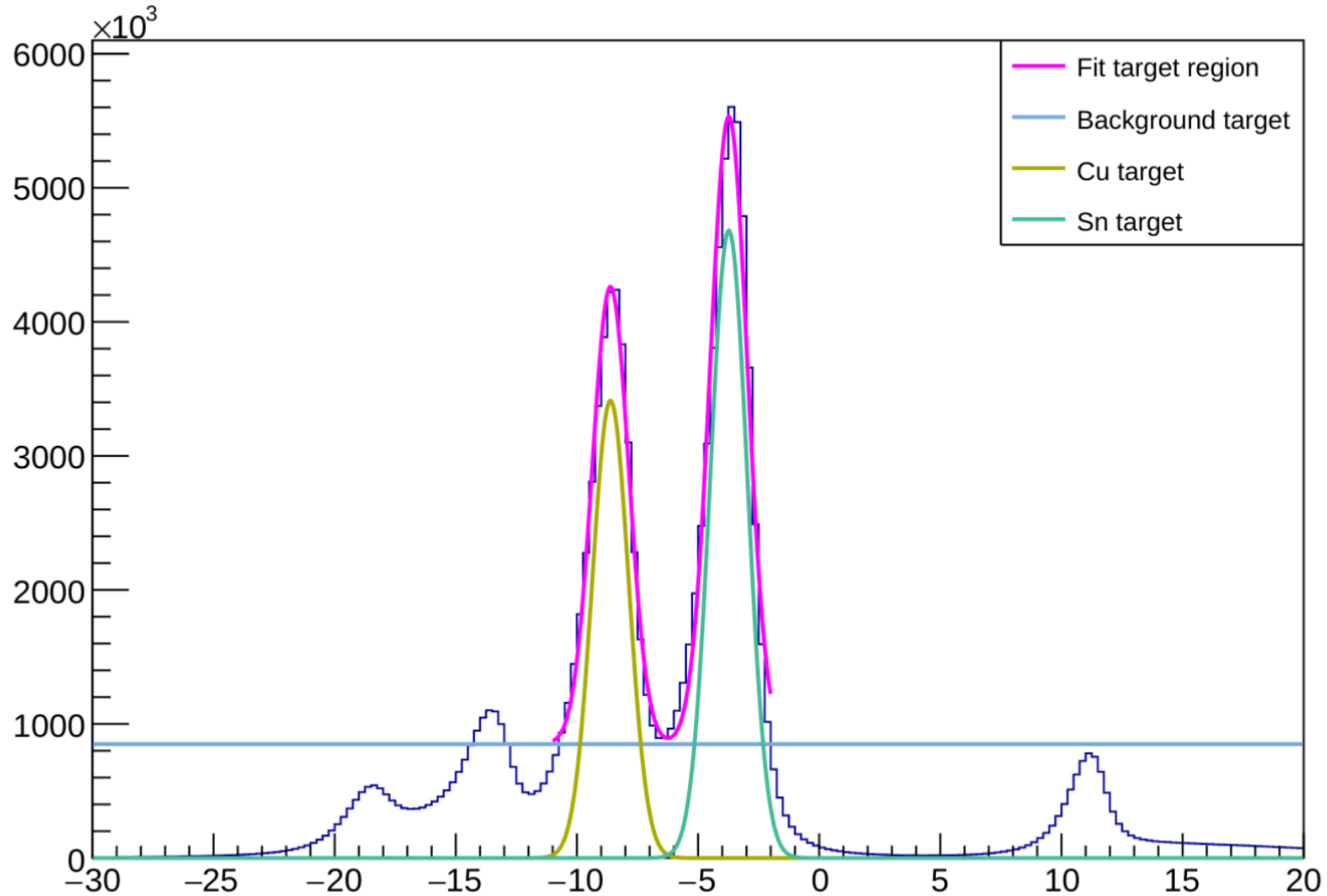
Fit target region with constant + gaus + gaus

RGD analysis: vz study LD2 region



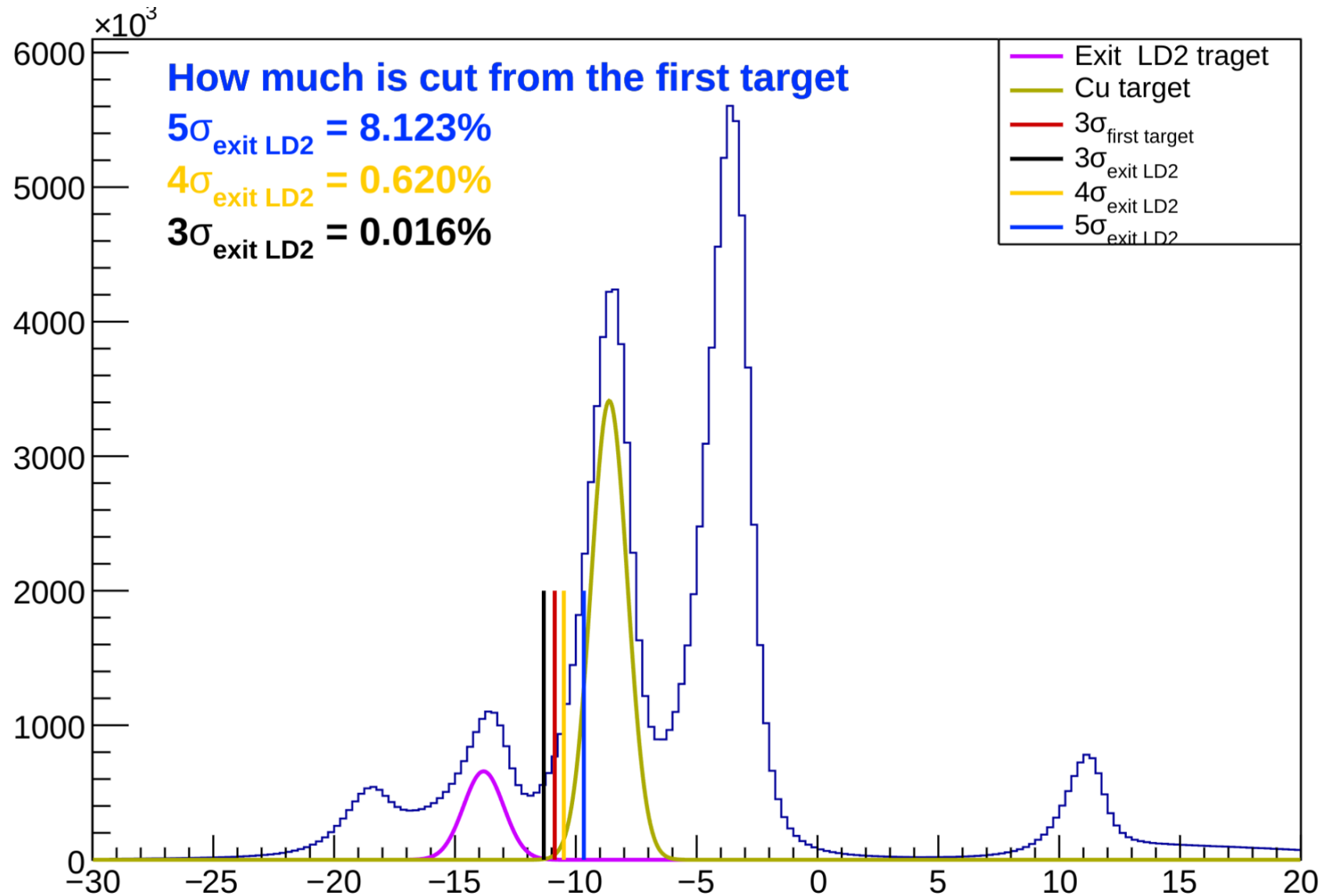
Fit decomposition for the LD2 region.

RGD analysis: vz study target region



Fit decomposition for the target region.

RGD analysis: vz study

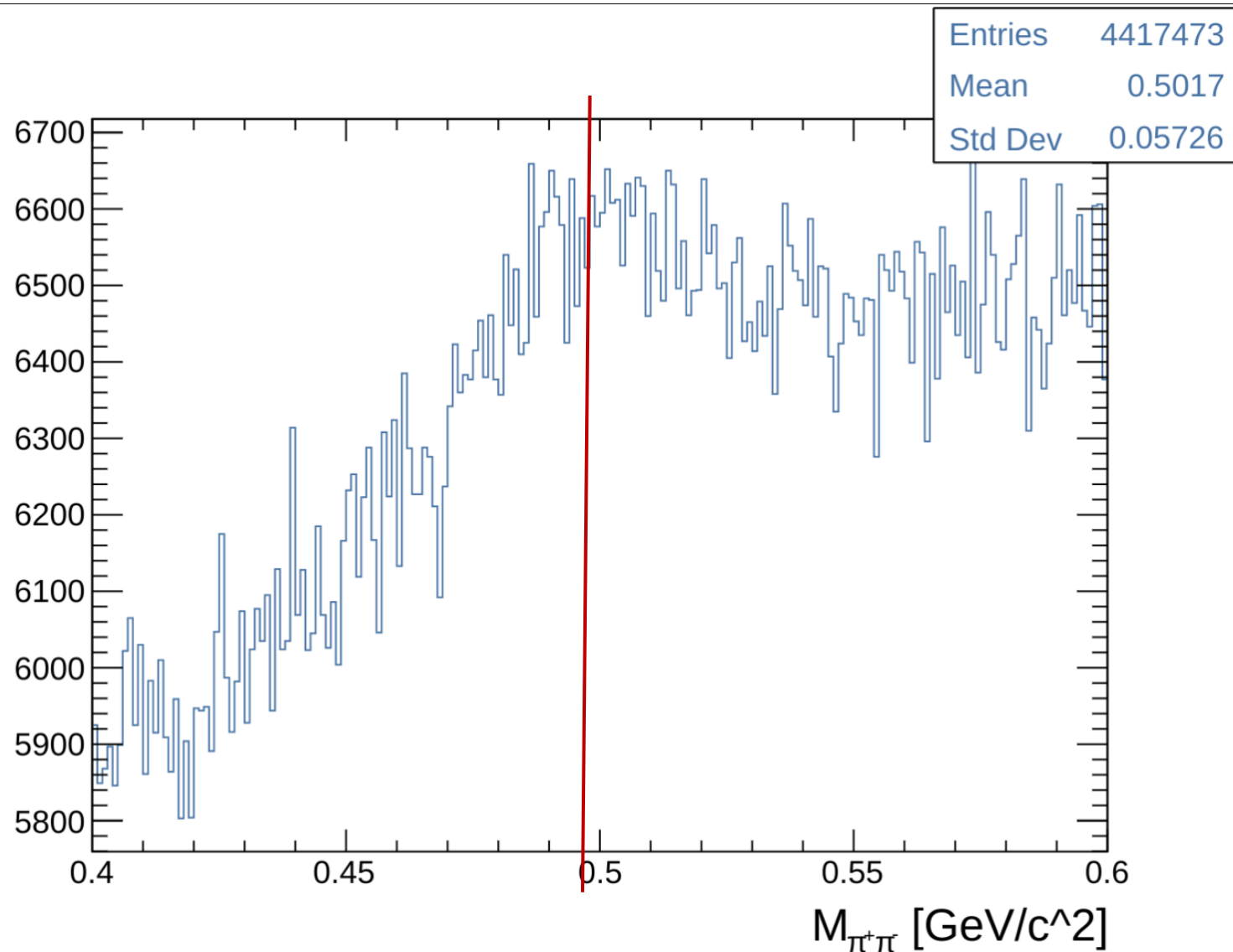


The blue line shows the 5σ from the Gaussian of the exit window LD2. If we apply the cut, the number of associates is the amount lost by the Gaussian of the first target.

The red line show the 3σ from the gaussian of the first target.

A cut at 3 or 4σ remove less than 1%.

RGD analysis: SIDIS K0 short



Electron cut:

- $-15 < v_z < 0 \text{ cm}$
- $-5 < \chi^2_{\text{pid}} < 5$
- $P > 1 \text{ GeV}/c$

Event:

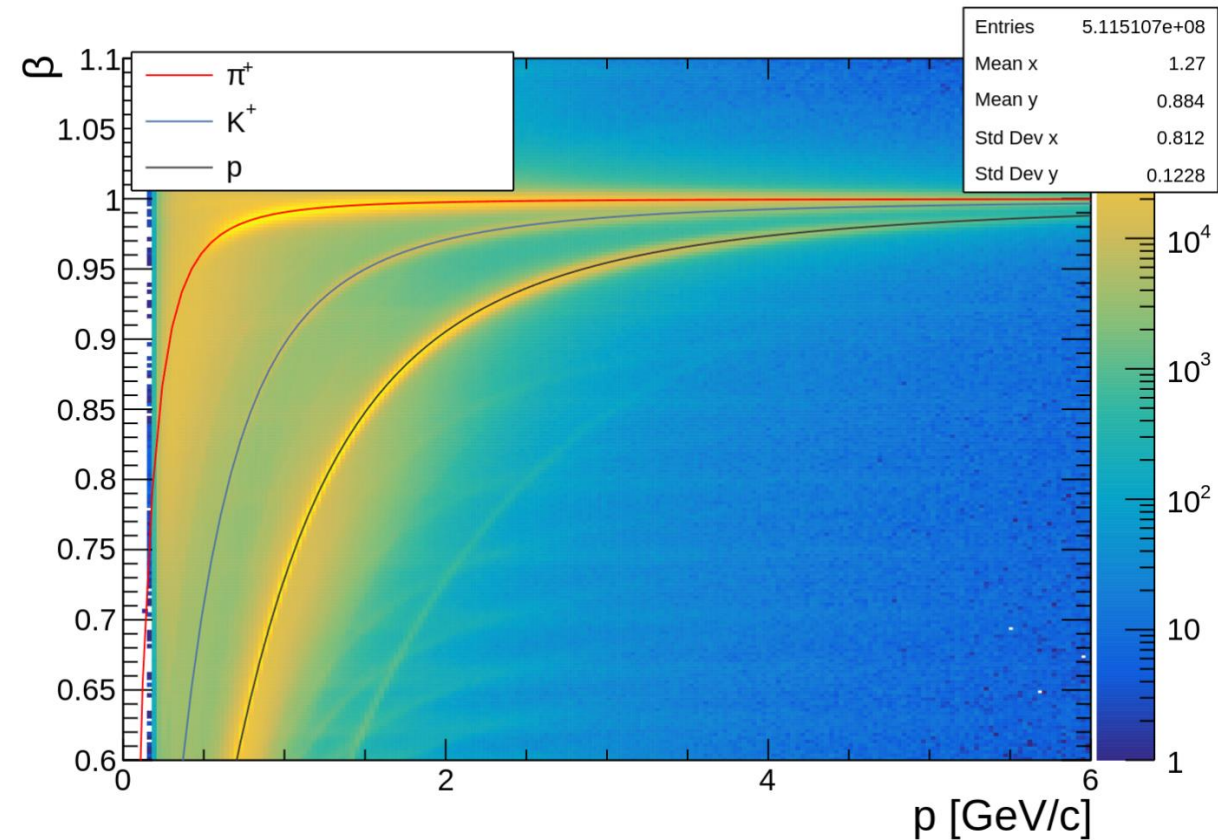
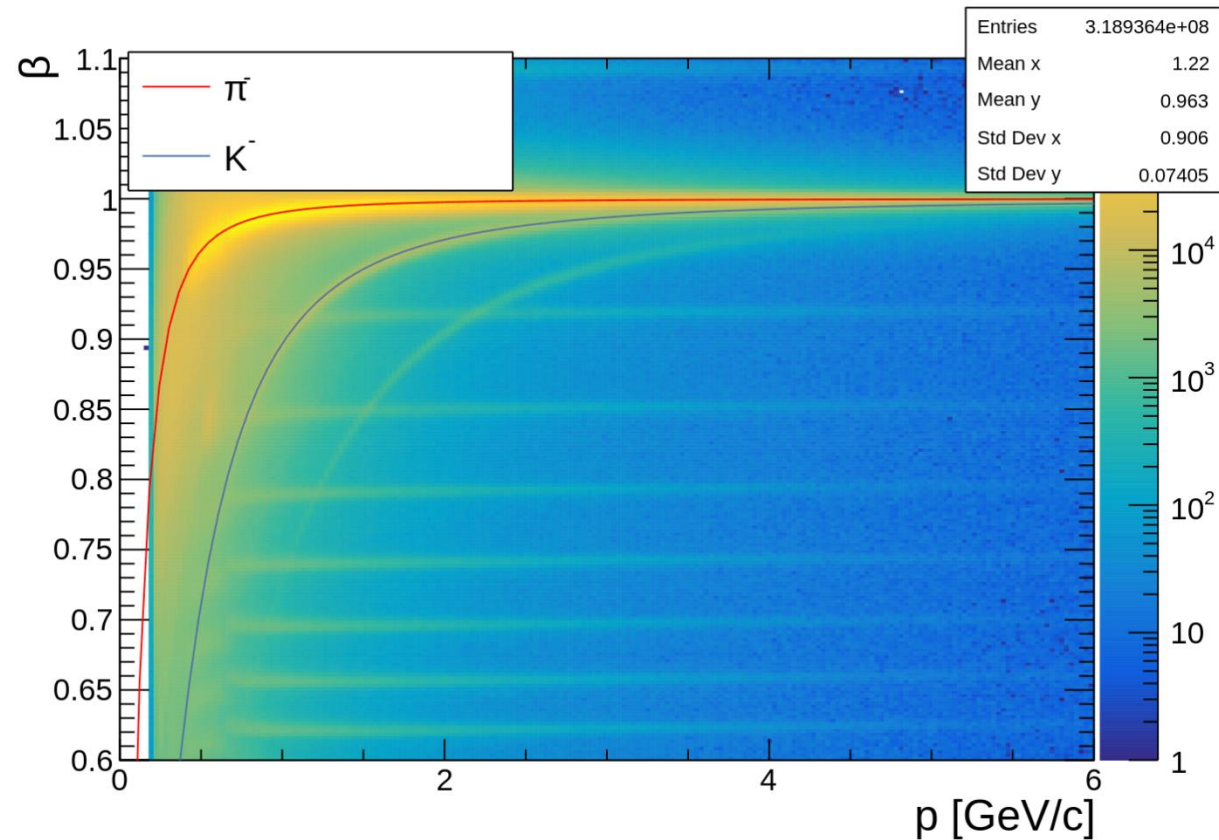
- $Q^2 > 1 \text{ GeV}^2$
- $y < 0.85$
- $W > 2 \text{ GeV}$

Pions:

- $P < 3 \text{ GeV}/c$
- $-3 < \chi^2_{\text{pid}} < 3$
- $Z_h < 0.7$

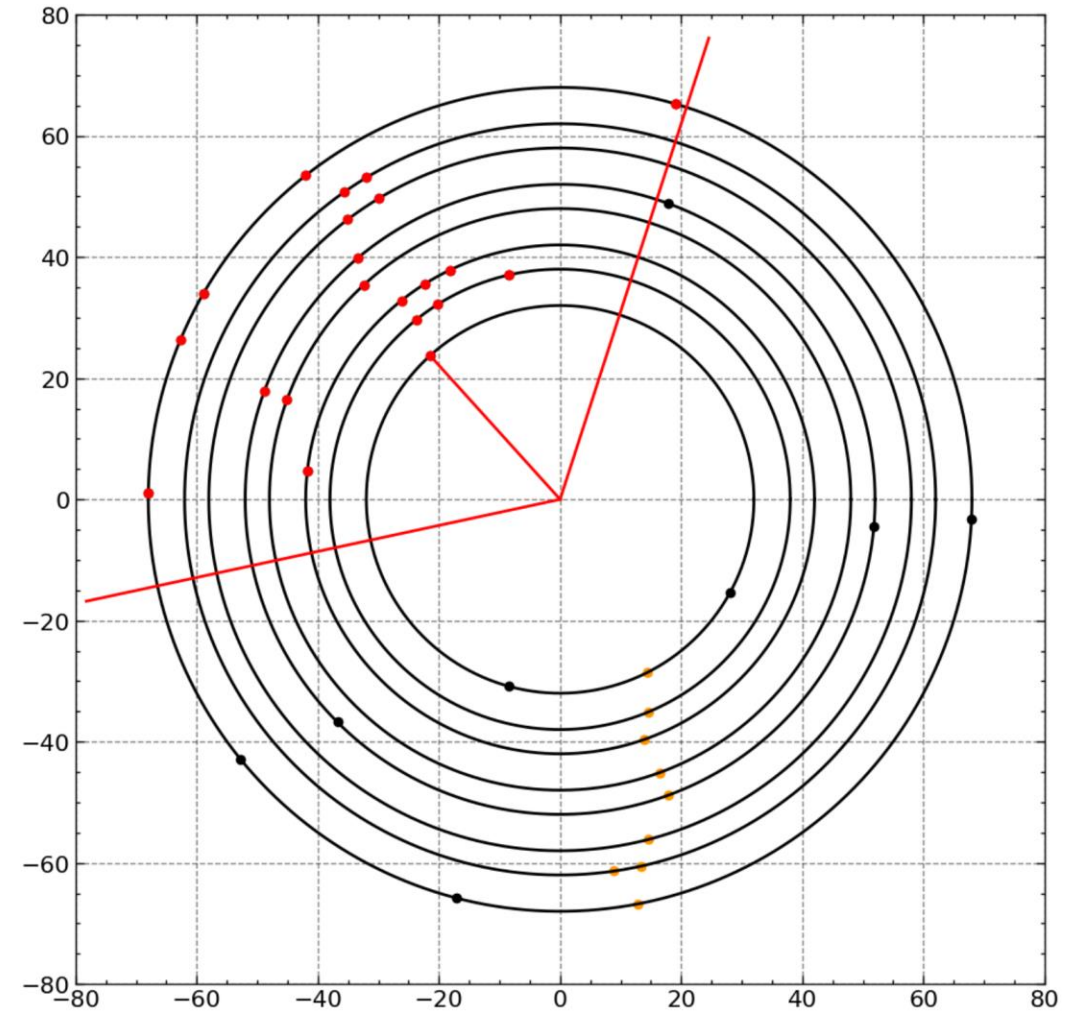
RGD analysis: SIDIS

- β vs p for both negatively and positively charged particles:



ALERT: AI for track finding

- Goal: reduce the number of possible tracks.
- Forcing track to start by a seed (precluster on the first layer)
- No more than 2 preclusters per layer.
- Use a pre-trackfinder based on the angle of the seed.
- Find track with 7, 8 or 9 preclusters.



ALERT: AI for track finding

- The mean of the possible tracks is 2500.
- But some events have a much higher number of tracks: up to 300K
- Inference time of the model is around $15\mu\text{s}$ (60KHz).
- So some event will take more than 5s (time to find all the track is large also)
- Possiblity: limit the tracks to the ones with 8 preclusters (1 on each layer).
- Or better solution: more clustering: merge preclusters if there are in the same superlayer.
 - Reduce the number of preclusters to 5.
 - We only need to find tracks with 5 preclusters.

To do list

- RG-D analysis:
 - Follow up on the cross check of Daniel analysis.
 - Understand how pid and chi2pid is assign to π/K
 - Multiplicity ratio for K^+/K^-
 - Better pid for K^+/K^-
- AI ALERT:
 - Check number of tracks with we use the other clustering
 - MLP efficiency as function of background occupancy / number of tracks.